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EE 472

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Homework #7

The screenshot displays the Quartus Prime IDE interface. On the left, the 'Task' pane lists various design steps, all of which are marked with green checkmarks, indicating successful completion. The main area shows a Verilog HDL code editor with a file named 'hw7'. The code defines a module 'hw7' with an output register 'z' and inputs 'clk', 'reset', and 'w'. It includes parameters 'A', 'B', and 'C' with binary values. The code is divided into two main blocks: 'STATE_MEMORY' and 'NEXT_STATE_LOGIC'. The 'STATE_MEMORY' block handles the state transition logic, and the 'NEXT_STATE_LOGIC' block handles the output logic. The code is as follows:

```
1 module hw7(output reg z,  
2     input wire clk, reset, w);  
3     reg current_state, next_state;  
4     parameter A = 2'b00,  
5               B = 2'b01,  
6               C = 2'b10;  
7  
8     always @ (posedge clk) //synchronous active high reset  
9     begin: STATE_MEMORY  
10        if (reset)  
11            current_state <= A;  
12        else  
13            current_state <= next_state;  
14        end  
15    always @ (current_state or w)  
16    begin: NEXT_STATE_LOGIC  
17        case (current_state)  
18            A : if (w==0)  
19                next_state = A;  
20            else  
21                next_state = B;  
22            B : if (w==0)  
23                next_state = A;  
24            else  
25                next_state = C;  
26            C : if (w==0)  
27                next_state = A;  
28            else  
29                next_state = C;  
30        endcase  
31    end  
32    always @ (current_state or w)  
33    begin: OUTPUT_LOGIC  
34        case (current_state)  
35            A : z = 0;  
36            B : if (w==0)  
37                z = 0;  
38            else  
39                z = 1;  
40            C : if (w==0)  
41                z = 0;  
42            else  
43                z = 1;  
44        endcase  
45    end  
46 endmodule  
47  
48  
49  
50
```

At the bottom, the 'Messages' pane shows a green message: '293000 Quartus Prime Full Compilation was successful. 0 errors, 19 warnings'.